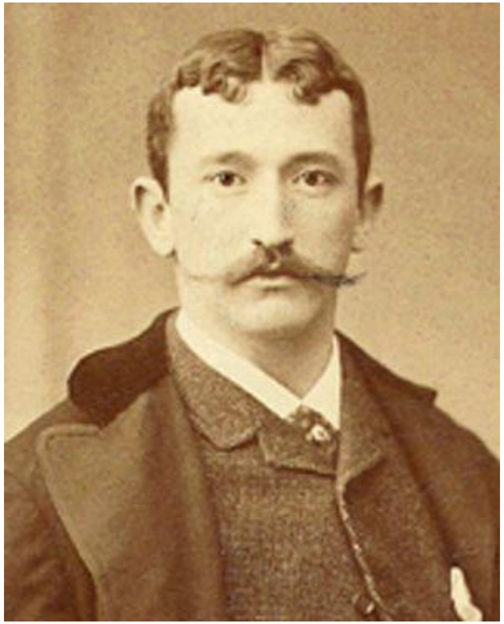
Approach to Regional Anesthesia

DR. AMIR M. BOUJAN

History

- The natives of Peru have chewed the leaves of the indigenous plant “Erythroxylon coca” the source of cocaine, to induce a feeling of well-being
- Koller used cocaine for the eye in 1884
- Halsted & Hall used cocaine as nerve block
- Spinal anesthesia was first performed by August Bier in Germany in 1889
- First synthetic local-- procaine in 1905
- Lidocaine synthesized in 1943



Regional anesthesia includes

1. Anesthesia:

- Spinal (for the abdomen, pelvis & lower limb)
- Epidural (for the neck, thoracic, abdomen, pelvis & lower limb)
- Caudal (for perineal surgery)

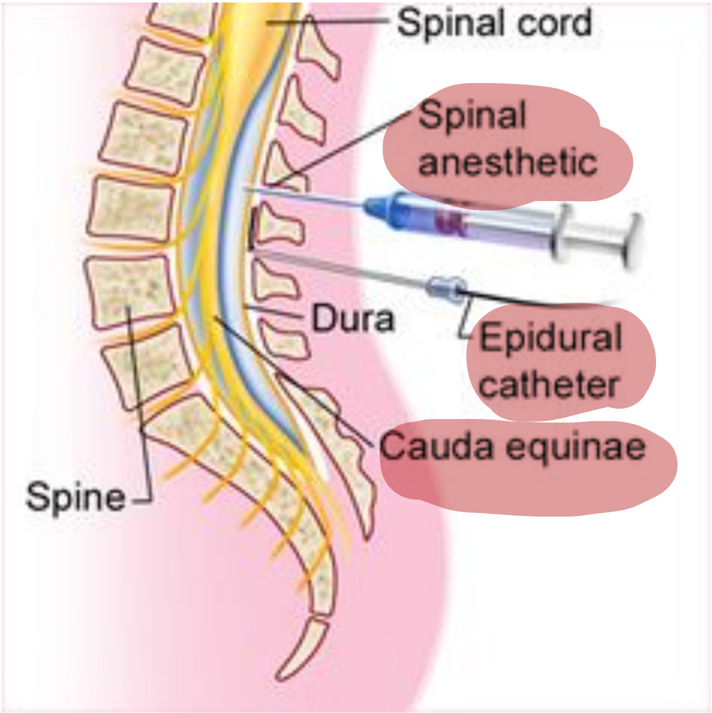
2. IV regional block (for upper & lower limb)

3. peripheral nerve blocks:

- Interscalene brachial plexus block (for shoulder and elbow surgery)
- Axillary, Supraclavicular, Infraclavicular brachial plexus block (for every other part of the arm)
- Sciatic nerve block (for almost all the leg)
- Femoral nerve block (for the rest of the leg)

Regional Anesthesia

With a cooperative patient, regional anesthesia may ensure the appropriate immobility and analgesia required for surgery, without exposing the patient to the risks of general anesthesia.



Regional Anesthesia

Superficial and deep operations on the extremities— particularly the distal extremities—may be amenable to peripheral nerve block Because surgical anesthesia may be achieved without sedation, this technique is particularly attractive in patients for whom systemic disease (e.g., severe pulmonary disease, cardiovascular disease, or renal failure) may present a significant challenge during general anesthesia.

Regional Anesthesia

- Regional anaesthesia is ideal for many operations, in particular those on the limbs and lower abdomen.
- For those who do not wish to be fully awake for surgery, sedation can also be used.
- For many other operations, regional analgesia can complement GA and provide lasting and effective post-operative pain relief.

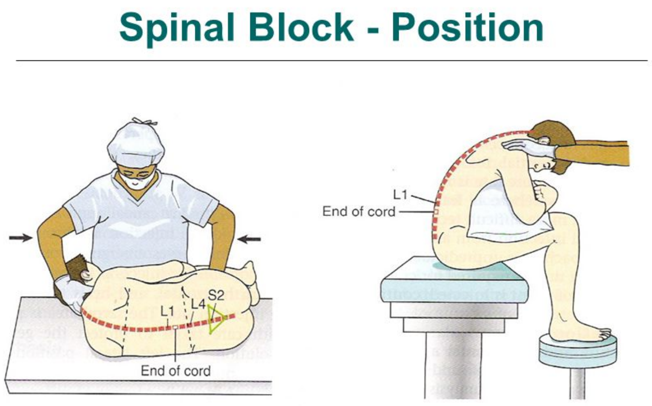


FIG. 24-15 Spinal block—lateral position.

FIG. 24-16 Spinal block—sitting position.

Regional Anesthesia

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- Although regional anaesthesia has side effects and complications, it is an excellent option for many patients, especially those with major co-morbidities such as significant heart and lung disease.
- Spinal and epidural anaesthesia are the techniques of choice for a Caesarean section for the vast majority of women.

Regional Anesthesia

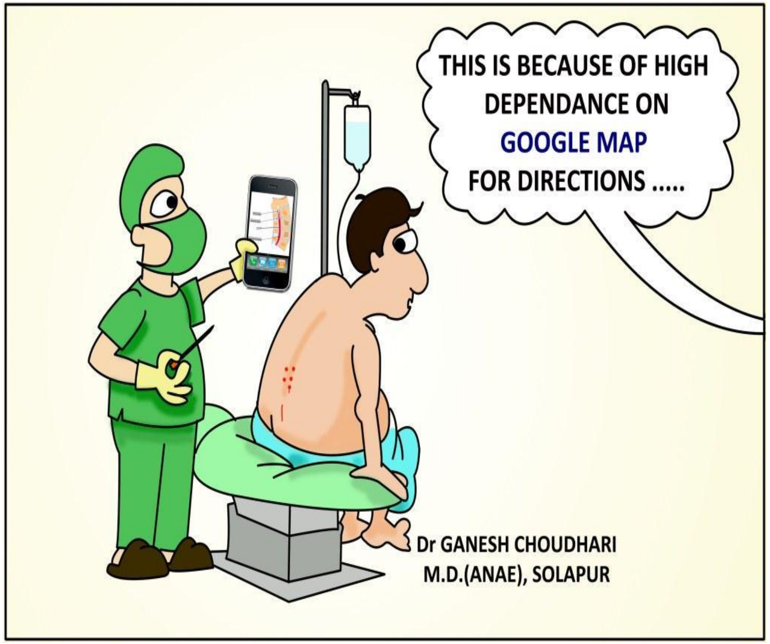
A detailed knowledge of the anatomy, good manual dexterity, current resuscitation skills, and the willingness to accept that they will need to be committed to deliver the best results are essential for the practitioner

Regional Anesthesia

Unfortunately, regional anaesthesia can never offer 100% success, and a ‘plan B’ is essential—this will usually be GA. The knowledge, skills, and facilities necessary to provide safe GA are therefore a prerequisite for the performance of regional anaesthesia

Safe Practice

- Patient consent and preparation
- Equipment
- Monitoring
- Assistance
- Environment
- Documentation



Contraindication for Neuraxial blocks

Relative contraindications:

- Specific cardiac disease (e.g. severe valvular stenosis, Eisenmenger's syndrome, peripartum cardiomyopathy).
- Hypovolaemia (hypotension)
- Expectation of significant haemorrhage
- Previous back surgery (technical difficulty)
- Neurological disease
- Systemic sepsis (increased incidence of epidural abscess, meningitis)

Absolute contraindications:

- Patient refusal
- Anticoagulation
- Local sepsis
- Allergy (true allergy to amide LAs is rare).
- Raised ICP

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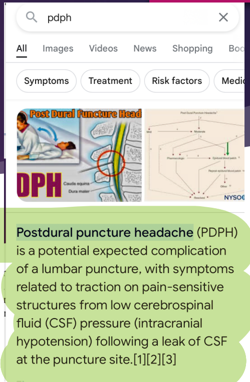
Complications of Neuraxial Blocks



- Hypotension
- Bradycardia (if block extends to the mid-thoracic region)—can progress to cardiac arrest.
- High block, compromising breathing, may extend to 'total spinal'.
- Urinary retention.
- Nerve damage

- PDPH
- Infection: abscess, meningitis.
- Bleeding: spinal canal haematoma—more likely in patients with disorders of coagulation. Can cause spinal cord and permanent paraplegia
- Serious complications (death or paraplegia of 1:50 000–1:140000)

نادرة

LA Systemic Toxicity (LAST)

Local anesthesia.

- Occurs when an excessive amount of LA enters the circulation.
- Toxicity can be avoided by injecting slowly and in small boluses, interspersed with frequent gentle aspirations to exclude accidental intravascular needle placement, especially when single-site large doses are administered.
- Maximum dosages vary, depending on the age and size of the patient, the site to be anaesthetized, vascularity of the tissues, individual tolerance, and the anaesthetic technique,

Presentation of LAST

Local anesthesia systemic toxicity

- Light-headedness, dizziness, drowsiness.
- Tingling around the lips, fingers, or generalized.
- [illegible]
- Metallic taste, tinnitus, blurred vision.
- Confusion, restlessness, incoherent speech, tremors, or twitching, leading to convulsions, with loss of consciousness and coma.
- Bradycardia, hypotension, CVS collapse, and respiratory arrest. ECG changes

Management of LAST

Local anesthesia systemic toxicity

- Discontinue injection.
- ABC … 100% O2 .
- Intubate and ventilate, if required, to prevent hypoxic CVS collapse. Hyperventilation may help by increasing the pH in the presence of metabolic acidosis.
- CPR if pulseless—commence the ALS protocol
- Treat convulsions with IV midazolam (3–10mg), diazepam (5–15mg),, propofol (20–60mg), or thiopental (50–150mg). Titrate against the patient response.
- Give an IV bolus injection of 20% lipid emulsion, e.g. Intralipid® 20%, 1.5mL/kg over 1 min (100mL for a 70kg patient). Then start infusion

Tx

Allergic reactions to LA

Local Anesthesia

- Allergic reactions to LAs are extremely rare. The ester groups are more prone to exhibit allergic reactions than amides.
- Allergic reactions range from simple local irritation with rash or urticaria to laryngeal oedema or anaphylaxis.

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Choice of Agent

Lidocaine

- Medium-acting amide. Moderate vasodilatation. Cerebral irritation before cardiac depression.
- Duration: enhanced and peak plasma levels reduced by adrenaline. Available in high concentrations for use in topical airway anaesthesia (4% solution and 10mg/dose spray).
- Dose 3mg/kg (7 with adrenaline)

Adrenaline

Bupivacaine

- Long-acting amide, used For long-lasting post-operative analgesia,
- Dose 2mg/kg
- Prolonged cardiotoxicity in higher doses

Why is lidocaine given with adrenaline?

Lidocaine and epinephrine combination injection is used to cause numbness or loss of feeling for patients having certain medical procedures (by blocking ...

May 1, 2024

https://www.mayoclinic.org › drg-...

Lidocaine And Epinephrine (Injection Route) Precautions - Mayo Clinic

Epinephrine (adrenaline) is usually injected with lidocaine to prolong the local anesthetic effect. Jun 24, 2021

https://www.ncbi.nlm.nih.gov › pmc

Incidence of the "Adrenaline Rush" and Vasovagal Response with ...

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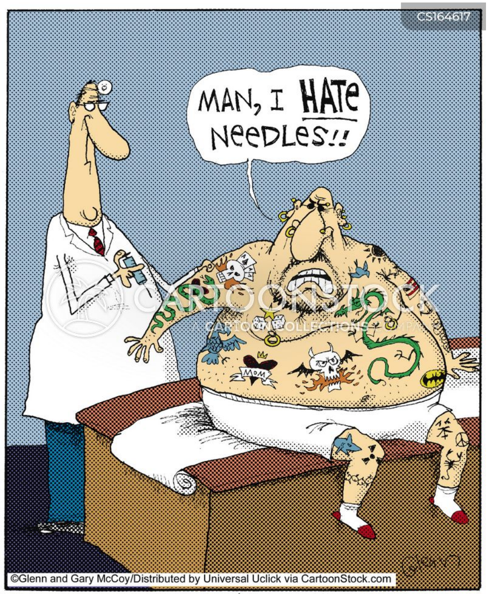
Addition of epinephrine to local anesthetic solutions

Adrenaline

Epinephrine 1:200000 or 5µg/ml

Advantage:

1. Limits systemic absorption
2. Prolongs the duration of local anesthetics by keeping them in contact with nerve fibers
3. Decreasing the possibility of systemic toxicity cause the rate of metabolism will match the rate of absorption
4. As test in epidural regional anesthesia



Addition of epinephrine to local anesthetics is not recommended in:

1. Unstable angina pectoris
2. Cardiac dysrhythmias
3. Uncontrolled hypertension
4. Uteroplacental insufficiency
5. Peripheral nerve block anesthesia in areas that may lack collateral blood flow (digits, penis)
   end artery
6. Intravenous regional anesthesia

→ لأنه رح IV استعمالها
Adrenaline ح تقلب
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References

- Basics of Anesthesia
- Oxford Handbook of Anesthesia


11:44 Pm.
Fri 7. Jun
Thank You